

Edward James Isoghie

PhD Candidate, Industrial & Systems Engineering, University of Louisville

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PROFESSIONAL SUMMARY

Human Factors Scientist with knowledge in human performance, work system design, safety analysis, mixed-method research, usability testing frameworks, and continuous process improvement. Experienced in applying human-centered design principles and ergonomic principles with a proven ability to analyze and synthesize data in order to generate insight and provide actionable recommendations to meet societal needs.

CORE COMPETENCIES

- Human Factors
- Hazard/Risk Assessment
- Human Subjects Study Design
- User Research
- Formative & Summative Usability Studies
- Programming (Python, R)
- Proficiency in Statistics and use of Statistics software (Excel, SPSS)
- Strong Communication and Interpersonal skills
- Workflow & Task Analysis
- HF Documentation for Regulatory Submissions

PROFESSIONAL EXPERIENCE

Human Factors Researcher | Center for Human Systems Engineering

University of Louisville | January 2025 – Present

- Conduct human factors research to identify user needs, use environments, and task flows for safety-critical systems.
- Perform task analysis, use error identification, and use-related risk assessment to inform design decisions.
- Plan and execute formative usability evaluations using interviews, scenario-based testing, and structured observation.

Achievements

- Conducted user research using **task analysis, structured interviews, and observational studies** to understand human-technology interaction in complex systems.
- Developed personas and workflow analyses to identify cognitive load, perceptual constraints, and potential use-related hazards.
- Synthesized qualitative and quantitative findings into structured usability requirements.
- Recruited and managed test participants; developed study protocols and test scripts.
- Performed root cause analyses for observed use errors and performance breakdowns.

Graduate Teaching Assistant – Center for Human Systems Engineering

University of Louisville | January 2024 – December 2024

- Supported courses in Safety Management, Facilities Design, Simulation, and Human-Centered Systems.
- Assisted with hazard analysis, human-system interaction evaluation, and workflow modeling in applied projects.

- Guided students in risk analysis, task decomposition, and usability evaluation methods relevant to medical and industrial systems.

Associate Technology Consultant (Digital Transformation) | Deloitte | February 2022 – June 2022

- Supported user-centered redesign of enterprise workflows across order-to-cash and procure-to-pay processes.
- Conducted stakeholder interviews, requirements elicitation, and workflow analysis to identify usability gaps.
- Documented as-is/to-be workflows, usability pain points, and system constraints for cross-functional teams.
- Collaborated with cross-functional engineering and business teams to analyze complex operational systems and identify human-process breakdowns affecting performance and safety.
- Delivered structured documentation and executive-ready presentations synthesizing large datasets into actionable insights.

Distribution Operations Specialist (Human Factors & Safety-Critical Operations)

Eko Electricity Distribution Company (EKEDC) | May 2021 – July 2022

- Supported operations in a safety-critical environment, analyzing human-system interactions affecting reliability and risk.
- Performed use-related failure analysis tied to operational workflows and maintenance activities.
- Developed dashboards to monitor human performance indicators, error trends, and corrective actions.
- Collaborated with engineering and field teams to reduce use-related errors through workflow and process improvements.
- Worked in a safety-critical infrastructure environment supporting procedural compliance, hazard mitigation, and operational reliability.
- Assisted in documentation and analysis of operational incidents and near-misses, identifying human and systemic contributors.
- Contributed to improving safety communication and task clarity to reduce risk exposure.

FAMILIAR STANDARDS & REGULATORY KNOWLEDGE

- FDA Guidance: Applying Human Factors and Usability Engineering to Medical Devices
- ISO 9241-210: Human-Centered Design

TOOLS & METHODS

- Quantitative & Qualitative Analysis (SPSS, Python, and R)
- Usability Testing (Formative & Summative)
- Interviews, Focus Groups, Scenario-Based Evaluation
- Task Analysis & Workflow Modeling
- Use Error Analysis

PROFESSIONAL AFFILIATIONS

- Human Factors and Ergonomics Society (HFES)
- Institute of Industrial & Systems Engineers (IISE)
- Association for the Advancement of Artificial Intelligence (AAAI)

EDUCATION

Doctor of Philosophy (Ph.D.) in Industrial & Systems Engineering, January 2024- Present
Center for Human Systems Engineering, University of Louisville, Kentucky, USA.

Master of Science (MSc) in Operations Management, December 2023
Manderson Graduate School of Business, The University of Alabama, Tuscaloosa, USA.

Master of Science (MSc) in Industrial and Production Engineering, January 2022
Faculty of Technology, The University of Ibadan, Ibadan, Nigeria.

Bachelor of Science (B.Sc.) in Industrial and Production Engineering, February 2018
Faculty of Technology, The University of Ibadan, Ibadan, Nigeria.

AWARDS AND HONORS

- **Human Factors and Ergonomics Society (HFES), Technical Training- Best Paper Award- June 2025**
Refereed recognition for “The Interplay between Situation Awareness (SA) and Cognitive Workload in Virtual Reality Training Environments: A Comprehensive Systematic Review”.
- **American Society for Engineering Education (ASEE) Industrial Engineering Division – Outstanding Paper Award- 2025**
Refereed recognition for The Use of Generative AI for the Rapid Development of Qualitative Interview Transcripts for a Human-Centered Design problem.
- **National Science Foundation (NSF) – Travel Grant - 2025**
Competitive support to present federally aligned research in human-centered systems and education.
- **Community-Engaged Scholarship (USD \$500) - 2025**
Equity-focused co-design with a health-equity organization for culturally responsive data collection.
- **University Research Mentorship Award -2024**
Departmental recognition for supervising undergraduate research and building capacity in applied projects.

PUBLICATIONS

Published Articles:

1. Isoghie, E.J, & Saleem, J.J (2025) Drivers and Barriers to High Situation Awareness in Manufacturing: A Literature Review. Conference Proceedings of Human Factors and Ergonomics Society (HFES) Published.
2. Isoghie, E.J, & Saleem, J.J (2025) The Interplay between Situation Awareness (SA) and Cognitive Workload in Virtual Reality Training Environments: A Comprehensive Systematic Review. Conference Proceedings of Human Factors and Ergonomics Society (HFES) Published.
3. Isoghie, E.J, Saleem, J.J, Tretter, T & Hieb J.L (2025). The Use of Generative AI for the Rapid Development of Qualitative Interview Transcripts for a Human-Centered Design problem. American Society for Engineering Education.
4. Saleem, J.J, Isoghie, E.J, Tretter, T & Hieb J.L (2025). Systematic Development of a Rubric for Assessing a Human-Centered Design Problem using a Tiered Framing of Depth for Student Thinking. American Society for Engineering Education.

Articles under Review

1. Isoghie, E.J, Saleem, J.J, Tretter, T & Hieb J.L (2026). An Examination of Knowledge and Effective Use of Mixed Methods in Industrial Engineering Practice. IISE Transactions (Under Review)
2. Isoghie, E.J & Saleem, J.J, (2026). Exploring the Contextual Application of Situation Awareness (SA) as a Non-Technical Skill in Connected Manufacturing. Conference Proceedings of Human Factors and

Ergonomics Society (HFES) (Under Review).

3. Isoghie, E.J & Saleem, J.J, (2026). Informing AI Prompt for Industrial Engineering Training: A Human Factors Engineering Course Approach. American Society for Engineering Education (Under Review).
4. Isoghie, E.J & Saleem, J.J (2026). Informing Engineering Education Through the Industry: A Survey of Situation Awareness in Manufacturing Practice. American Society for Engineering Education (Under Review).
5. Isoghie, E.J, Saleem, J.J, Tretter, T & Hieb J.L (2026). Introducing a Mixed-Methods Approach to Engineering Students through Human-Centered Design. 2026 NSF Grantees Poster Session, American Society for Engineering Education.

Work in Progress Paper

1. Isoghie, E.J, Saleem, J.J, Tretter, T & Hieb J.L (2026). **Scenarios and Education Needs of Mixed Methods in Engineering Practice. To be submitted to Journal of Engineering Education**
2. Isoghie, E.J & Saleem, J.J, (2026). Situation Awareness in Manufacturing: **A literature Review. To be submitted to IISE Transactions on Occupational Ergonomics and Human Factors**

CONFERENCE / INVITED PRESENTATIONS

- 10/13/25 Drivers and Barriers to High Situation Awareness in Manufacturing: A Literature Review. (Paper Podium Presentation). 2025 Human Factors and Ergonomics Society (HFES) Annual Conference & Exposition, Chicago, Illinois.
- 10/13/25 The Interplay between Situation Awareness (SA) and Cognitive Workload in Virtual Reality Training Environments: A Comprehensive Systematic Review. (Paper Podium Presentation). 2025 Human Factors and Ergonomics Society (HFES) Annual Conference & Exposition, Chicago, Illinois.
- 06/23/25 The Use of Generative AI for the Rapid Development of Qualitative Interview Transcripts for Human-Centered Design Problems. (Paper Podium Presentation). 2025 American Society for Engineering Education (ASEE) Annual Conference & Exposition, Montreal, CA.